

AgamBhitt: The Autonomous Cyber Security Overseer

An Agentic "Self-Healing" Immune System for Financial Infrastructure

AgamBhitt (Sanskrit for *Impenetrable Foundation*) is an autonomous cybersecurity orchestrator designed to transform reactive banking security into a proactive, reasoning-based defense layer. By fusing Non-Euclidean Graph Intelligence with Generative AI, AgamBhitt creates a "Digital Twin" of a bank's security posture to predict attack vectors and automate remediation in milliseconds.

The Tri-Layered Defense Architecture

- The Wall (Contextual Edge Protection):** A Transformer-based WAF utilizing fine-tuned RoBERTa/DistilBERT models. Unlike legacy regex filters, it analyzes the semantic intent of traffic to neutralize zero-day exploits and polymorphic payloads.
- The Eyes (Predictive Topology Intelligence):** Leverages Graph Neural Networks (GNN) to map the bank's network of users, assets, and permissions into a Heterogeneous Graph. Link Prediction identifies potential paths to critical assets like SWIFT gateways before a breach occurs.
- The Brain (Agentic Orchestration):** An LLM-driven Overseer (Llama 3/Mistral) hosted on-premise. It ingests GNN alerts to generate CACAO-compliant playbooks and triggers Ansible/Terraform scripts for instant node isolation or configuration patching.

Continuous Evolution: The Self-Healing Loop

AgamBhitt differentiates itself through a 24/7 feedback loop between offensive and defensive agents:

- Offensive AI:** An integrated Red Team agent autonomously simulates attacks within a sandboxed Cyber-Range to identify hidden vulnerabilities.
- Defensive Feedback:** Upon finding a gap, the LLM generates a patch while the GNN re-tunes the WAF's focus, ensuring the infrastructure hardens itself against the newly discovered vector.

Technical Foundation

Layer	Core Technologies
AI/ML Engine	PyTorch Geometric, HuggingFace, vLLM
Data & Storage	PostgreSQL + pgvector, Neo4j
Action & Middleware	FastAPI, Redis, Ansible, Terraform
Visualization	SvelteKit, D3.js

Strategic Business Impact

- Zero-Day Resilience:** Detects novel attack patterns through latent representation learning rather than relying on outdated signature databases.
- Rapid MTTR:** Collapses Mean Time to Respond (MTTR) from hours to seconds via autonomous, agent-led playbook execution.
- Data Sovereignty:** Designed for strictly on-premise deployment, ensuring sensitive metadata and PII remain within the secure VPC to comply with RBI, DORA, and SEC regulations.

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